

Vision Transformer based Multi-Class Eye Disease Classification using Retinal Fundus Images

Minu Inba Shanthini Watson Benjamin
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
Chennai, India
minuben19@gmail.com

Kalyan Babu Gutti
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology
Chennai, India
kalyangutti19@gmail.com

Akash Maram
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D
Institute of Science and Technology
Chennai, India
vtu24837@veltech.edu.in

Ravikumar.S
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
Chennai, India
ravikumars@veltech.edu.in

Krishnaveni.N
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
Chennai, India
nkrishnaveni984@gmail.com

K. Maithili
Department of Computer Science &
Engineering
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
Chennai, India
maithilimtech@gmail.com

Abstract— In order to promote population-scale ophthalmic screening and prevent permanent vision loss, early and precise diagnosis of retinal disorders is essential to modern healthcare. Conventional diagnostic methods rely on ophthalmologists' manual assessment, which is laborious, subjective, and challenging to scale. The current study uses the Vision Transformer (ViT), an advanced deep learning architecture based on self-attention mechanisms, for multi-class classification and interpretability of retinal illnesses from fundus images in order to overcome these limitations. Ten retinal conditions can be identified by the suggested system including Diabetic Retinopathy, Glaucoma, Disc Edema, Retinal Detachment, and Retinitis Pigmentosa—and integrates Grad-CAM visualization to improve interpretability and strengthen clinical confidence. According to experimental data, the ViT model demonstrated great learning efficiency and generalization across several disease classes, achieving training accuracy of 93.21% with a loss of 0.1060 and validation accuracy of 78.91% with a loss of 0.9607. FastAPI, a lightweight, asynchronous, and scalable web interface that provides real-time predictions combined with Grad-CAM-based visual explanations for clinical validation, is used to implement the model in order to guarantee its practicality.

Keywords – Vision Transformer (ViT), Fundus Image Classification, Retinal Disease Detection, FastAPI Deployment, Grad-CAM Visualization, Explainable Artificial Intelligence (XAI).

I. INTRODUCTION

As people become more aware of their wellness today., tracking calorie intake has become a priority for many individuals. Accurate tracking of calorie intake is useful for people with obesity and diabetes and also for the prevention of diabetes and heart disease. Existing methods rely on Convolutional Neural Networks (CNNs) which is used for analysis of food from the photos but face challenges in feature

selection and generalization.

Traditional ophthalmic diagnosis methods rely on manual examination of fundus images by ophthalmologists. Although efficient, this method is subjective, time-consuming, and unsuitable for extensive screening initiatives. Automated systems that can effectively interpret medical images have been made possible by recent developments in deep learning. CNNs have demonstrated notable success in the analysis of retinal diseases and have been widely used in medical image classification. Nevertheless, the local receptive fields of CNN-based models limit their capacity to extract global contextual information from an image. Because of this, these models frequently have trouble differentiating between visually identical retinal diseases and lack interpretability, which is crucial for clinical applications.

In order to get over these restrictions, this work uses a cutting-edge deep learning model called the ViT architecture, which extracts both local and global characteristics from retinal fundus images by using self-attention mechanisms. Ten retinal conditions, such as diabetic retinopathy, glaucoma, disc edema, retinal detachment, and retinitis pigmentosa, can be categorized by the suggested system. Furthermore, to improve transparency and clinical trust, Gradient weighted Class Activation Mapping (Grad-CAM) is used to produce visual explanations that highlight the region's most important in the model's decision-making process.

To ensure real-world applicability, the trained model is deployed using FastAPI, a high-performance and asynchronous Python web framework that facilitates scalable, real-time disease prediction and visualization. This integrated approach provides an efficient and interpretable AI-based screening system, offering ophthalmologists an assistive tool to improve diagnostic accuracy and accelerate early detection of retinal disorders.

II. LITERATURE REVIEW

Deep learning, particularly transformer-based architectures, is gaining significant attention for its ability to enhance medical image analysis by improving diagnostic accuracy and interpretability. The following research demonstrate how sophisticated vision models, such the ViT, can accurately and scalably identify and categorize retinal disorders from fundus images.

The study [1] proposed a CNN-SVM hybrid model for retinal illness classification. The CNN extracted important spatial information from fundus images, whereas the SVM classified different retinal diseases. With an accuracy of over 93% when compared to a conventional CNN with Softmax. The model had high sensitivity and accuracy of better than 93% when compared to a conventional CNN with Softmax. The results of the research demonstrate that the hybrid architecture is efficient lessen the misclassification and can be used with automated retina. screening.

Niloy et al. [2] developed MobileNet-Eye, a transfer learning approach for efficient eye illness classification. The researchers tested MobileNetV2, ResNet50, and EfficientNet-B7 in a data set of 3,000 images of uveitis, eyelid conditions, and healthy eyes. MobileNetV2 had the highest accuracy of 96%, followed by EfficientNet-B7 with 95% and ResNet50 with 94%. The results impede that the lightweight designs are able to provide acceptable diagnostic accuracy while maintaining computational economy for real-time screening.

Purba et al. [3] tested how well the Vision Transformer design achieved in the ability to classify cataracts. The ViT model has used the self-attention mechanism for learning global picture associations making use of 1,120 samples of the retina under six experimental conditions. It received a maximum accuracy of 92.86 percent and a maximum precision of 100 percent, a maximum recall of 85.72 percent, and the F1-score of 92.31 percent. The research result showed that when it comes to diagnose cataract, Vision Transformers perform better than traditional CNNs.

Geethika et al. [4] conducted a study evaluating EfficientNet-B3, and ResNet-50 for automated classification of eye diseases. EfficientNet-B3 demonstrated increased accuracy and reliability, emphasizing the necessity of choosing the appropriate neural network architecture for ophthalmology processing of images.

Varghese and Pandian [5] developed a system for classifying eye illnesses using Inception-ResNetV2 with fine-tuning. After fine-tuning, accuracy improved from 81% to 94.74%, with major improvements on precision, recall, and F1-score. The study demonstrated how fine-tuning deep CNN models can increase diagnostic reliability for early diagnosis of eye disorders.

Giat et al. [6] assessed CNN models for automated categorization of eye diseases, including ResNet50 and VGG16. CNNs' potential for telemedicine and computer-aided ophthalmic diagnosis was confirmed by both designs' high accuracy under standardized review. The study demonstrated how crucial it is to select the appropriate CNN backbone for disease classification.

Shah et al. [7] examined a multimodal deep learning approach for Alzheimer's disease and mild cognitive impairment (MCI) early detection. Eye-gaze characteristics such pupil fluctuation, saccades, fixation time, and audio speech aspects were combined by the system. Combining modalities enhances early screening, as seen by the model's 79.2 percent accuracy for MCI and 82 percent accuracy for Alzheimer's disease.

Zain et al. [8] suggested using the ODIR dataset to develop a CNN-based system for multi-label eye illness detection. Data quality was enhanced by preprocessing using CLAHE and cropping. With an F1-score of 88.05 percent, an AUC of 87.77 percent, and a Cohen's Kappa score of 44.37 percent, MobileNetV2 outperformed ResNet152 and VGG19. The efficacy of lightweight CNNs for clinical screening was validated by the study.

Kumar et al. [9] investigated many pretrained models for the identification of eye diseases and added a majority-voting ensemble to increase robustness. 81.25 percent recall, 83.68 percent precision, 95.17 percent accuracy, and a 79.12 percent F1-score were all attained by the group. The findings showed that ensemble learning improves eye disease screening systems' dependability.

Orlosky et al. [10] investigated the use of eye-tracked virtual reality for Parkinson's disease remote diagnosis. Using an infrared eye-tracking camera, a virtual reality interface replicated clinical eye-movement tasks like smooth pursuit and fixation. Virtual reality tracking of eyes has been found to facilitate remote neurological assessment in both Parkinson's disease sufferers and healthy controls.

Wardani et al. [11] presented a hybrid classification strategy was proposed to improve the diagnosis of eye illnesses by combining Support Vector Machine (SVM) and K-Nearest Neighbor (KNN). The hybrid SVM-KNN model outperformed the standalone SVM with a 94.67 percent accuracy. The results demonstrated that merging multiple machine learning classifiers enhances screening reliability.

Li et al. [12] introduced a two-mode strategy that integrates Deep-learning-based RGB and infrared to detect ocular imaging surface diseases. Accuracy of the system in classifying. illnesses were 98.7 where F1-score was 0.980 identifying subconjunctival hemorrhages. It was 96.2, and the F1-score is less than 1% of 0.956. The system facilitated meibomian gland as well malfunction study and fitted a handheld infrared camera for home monitoring.

Shinde et al. [13] presented a deep learning model that employs EfficientNet-B3 and transfer learning and data augmentation to identify eye diseases. The model performed better than traditional CNNs having an accuracy of 97 and 95 on training and test data respectively. The study highlighted the usefulness and accuracy of EfficientNet-B3 in automated diagnosis.

Huong et al. [14] developed a YOLO algorithm with 1,500 photos of ill and healthy eyes. Applying YOLO to a Raspberry Pi 4 for real-time diagnostics resulted in over 94% recognition efficiency, highlighting its effectiveness for speedy and accurate embedded screening systems.

Kumar et al. [15] presented an unsteady recurrent autoencoder framework to detect the presence of strep throat infection early, and prevent acute rheumatic fever. This model integrates Denoised Variational Stacked Autoencoders (DVSE), nonlinear regression and Long Short-Term Memory networks to evaluate complicated and heterogeneous clinical data, such as throat images and blood serum samples. The system was found to be more robust with noisy and ambiguous datasets using multimodal medical data and recurrent feature learning. An experimental analysis showed that the proposed approach was more effective than the current approaches such as Blood Serum Content Analysis (BSCA), Smartphone-based Strep Throat Detection (SSTD), and Logistic Regression-based RF Detection (LRRF) about 10-15 percent. The paper provided evidence of the usefulness of nonlinear recurrent encoding and multimodal data fusing in the initial diagnosis of infectious diseases.

III. PROPOSED WORK

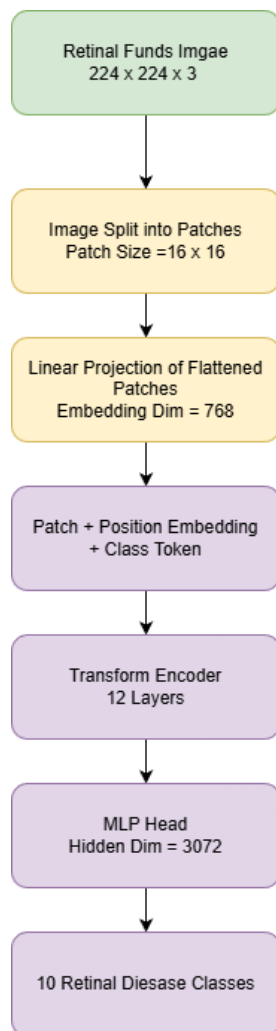


Fig. 1. Architecture of the Vision Transformer (ViT) Model Used for Multi-Class Eye Disease Classification

Fig. 1. Represents the overall workflow of the proposed Vision Transformer (ViT) architecture consists of the following major stages:

- i. **Input Image:** A retinal fundus image with a resolution of $224 \times 224 \times 3$ is provided as the input to the system.

- ii. **Patch Extraction:** The input image is divided into non-overlapping patches of size 16×16 pixels, resulting in a total of 196 patches.

- iii. **Patch Embedding:** Each extracted patch is flattened and projected into a 768-dimensional feature vector through a linear embedding layer.

- iv. **Token Formation:** Positional embeddings and a learnable class token are added to the patch embeddings to retain spatial information and enable global context representation.

- v. **Transformer Encoding:** The final token sequence is passed through 12 stacked transformer encoder layers, each consisting of multi-head self-attention, layer normalization, and a feed-forward neural network.

- vi. **Classification Head:** The encoded class token is forwarded to an MLP head with a hidden dimension of 3072, which performs the final feature transformation.

- vii. **Output:** The model generates a prediction by assigning the input fundus image to one of the ten retinal disease categories.

Dataset: The proposed retinal illness classification methodology relies on a combined dataset of retinal fundus images from the Original Eye illness Image Dataset (Mendeley) and the ODIR-5K (Ocular Disease Intelligent Recognition). Both datasets provide fundus images covering various retinal disorders and are widely utilized in ophthalmic research. Images from 10 preset retinal classes—CSCRCOLOR Fundus, Diabetic Retinopathy, Disc Edema, Glaucoma, Healthy, Macular Scar, Myopia, Pterygium, Retinal Detachment, and Retinitis Pigmentosa—are included in the combined dataset. The integration of images from several sources increases dataset diversity and strengthens the robustness of the proposed model by incorporating changes in imaging devices, illumination settings, and disease presentations. Only images with identifiable clinical traits and clear class definitions are retained in order to provide reliable learning and assessment. To protect data quality, photos with severe artifacts, low contrast, or indistinct visual patterns are removed during dataset preparation.

Pre Processing:

Preprocessing, an essential stage in machine learning, requires cleaning, transforming, and standardized raw data to improve its quality and consistency. It improves model performance by addressing a variety of data-related issues. The steps in processing are as follows. **Image Resizing and Normalization:** In order to satisfy the input needs of the ViT model, all retinal fundus images are reduced to 224×224 pixels. Images are transformed into tensor representations in RGB color space for deep neural network processing. It makes sure all samples have equal patch extraction and stable attention computation. **Data Augmentation,** Data augmentation techniques are exclusively used on the training set in order to improve robustness and decrease overfitting. These enhancements include brightness and contrast modifications, random rotation within a constrained angular range, and random horizontal flipping. **Dataset Splitting,** Following preprocessing, an 80:20 split ratio is used to separate the dataset into training and validation

subsets. To guarantee that the validation set is not viewed at all during training, the split is carried out at the dataset level. To allow for objective performance evaluation, data augmentation is strictly prohibited for the validation set. Encoding Categorical Variables, The Vision Transformer model receives the pre-processed images directly and internally splits each image into 16×16 pixel non-overlapping patches. The transformer encoder layers handle each patch once it has been flattened and projected into a high-dimensional embedding space. The model can capture both local retinal structures and broad contextual relationships thanks to its patch-based representation. The ViT architecture, which uses self-attention mechanisms to describe global contextual relationships in retinal fundus images, is the foundation of the suggested framework for classifying retinal diseases. The following steps make up the proposed system's overall workflow, which is depicted in Fig. 1.

Input Image Representation:

To ensure compatibility with the Vision Transformer input requirements, each retinal fundus image is scaled to a fixed resolution of $224 \times 224 \times 3$. Stable training across all samples and consistent patch production are made possible by the constant input size.

Patch Extraction:

Various models have been trained using extracted properties, such as shape, colour, texture, and nutritional information like proteins, fats, carbohydrates, and proteins. These models consist of.

Patch Embedding:

After being flattened, each extracted patch is mapped onto a 768-dimensional embedding space using a linear projection layer. In order to prepare the image patches for transformer-based processing, this phase converts them into a series of token embeddings.

Token Formation with Positional Encoding:

Each patch embedding is supplemented by learnable positional embeddings to retain spatial information lost during patch flattening. The token sequence also includes a learnable class token. This class token acts as the representative feature for the final classification by combining global data from all patches.

Transformer Encoder The token sequence is processed by 12 stacked transformer encoder layers, each comprising, Multi-head self-attention for modeling long-range dependencies across retinal regions, Feed-forward neural networks for non-linear feature transformation, Residual connections and layer normalization for training stability

Classification Head:

A multi-layer perceptron (MLP) head with a hidden dimension of 3072 receives the final representation of the class token. Ten classes of retinal diseases are represented by logits produced by the MLP head, and class probabilities are produced by using a softmax function.

Output Prediction:

CSCR_Color Fundus, Diabetic Retinopathy, Disc

Edema, Glaucoma, Healthy, Macular Scar, Myopia, Pterygium, Retinal Detachment, and Retinitis Pigmentosa are the categories in which the model predicts the class label.

Convolutional Neural Network:

Four convolutional blocks make up the sequential architecture used in the design of the custom CNN. To gradually extract hierarchical spatial information from retinal fundus images, each block consists of a convolutional layer, batch normalization, and max-pooling. The number of filters is increased to capture increasingly intricate retinal features as the network depth rises.

InceptionV3:

Using parallel convolutional paths with different kernel sizes, InceptionV3 is used as a deep transfer-learning architecture to capture multi-scale visual features. Because its initial classification head is removed, the pretrained InceptionV3 backbone can function as a potent feature extractor. A global average pooling layer is used to process the retrieved feature maps, and then fully connected layers with dropout are used to refine the features. Class probabilities for multi-class retinal disease classification are generated by the final softmax layer. Factorized convolutions are used in the architecture to strike a compromise between computational efficiency and representational capability.

MobileNetV2

A lightweight convolutional neural network designed for effectiveness and real-time inference is called MobileNetV2. It uses inverted residual blocks and depthwise separable convolutions to drastically cut the number of parameters without sacrificing performance. This work uses frozen convolutional layers on a pretrained MobileNetV2 backbone. Compact feature representations are extracted using global average pooling, and then fully connected layers with dropout regularization are used. The model can be used in contexts with limited resources because it produces predictions for ten different kinds of retinal diseases.

ResNet50

In order to address vanishing gradient problems and facilitate the training of deeper topologies, ResNet50 is a deep residual network that incorporates skip connections. It can learn high-level hierarchical features from retinal pictures by using the pretrained ResNet50 backbone without its classification head. The feature maps are subjected to global average pooling and then fully connected layers with dropout regularization. ResNet50 needs careful fine-tuning and enough data to perform well in challenging medical picture classification tasks, despite its architectural depth and residual learning capability.

A final loss of 0.6790 was attained by the custom CNN model. The model's ability to learn basic spatial information from retinal fundus images is demonstrated by the comparatively moderate loss value. The observed loss, however, points to limitations in simulating intricate and subtle clinical changes across several types of retinal diseases, especially when differentiating cases that are visually identical.

With a final loss of 0.5760, InceptionV3 outperformed the baseline CNN. This decrease in loss is a result of the multi-scale convolutional design of the model, which improves feature representation. The loss value shows

that even this improvement, the model still struggles to achieve stable convergence across all disease categories due to intra-class variation and overlapping visual characteristics.

The final loss for MobileNetV2 was 0.6910. Similar learning behaviour is indicated by the loss value being similar to that of the baseline CNN. Although MobileNetV2 has a lightweight and effective design, it is unable to capture complicated retinal structures due to its limited model capacity, which increases residual error during optimization.

With a final loss of 1.5667, ResNet50 showed the largest loss of all the models that were examined. The deep residual architecture's poor convergence and inadequate adaptation to the retinal illness classification task under the specified training configuration are indicated by this high loss value. The outcome implies that in order to obtain stable learning in multi-class ophthalmic applications, deeper residual networks need more thorough fine-tuning or larger datasets.

Out of all the models that were assessed, ViT showed the best learning behaviour. According to experimental results, the ViT model obtained a validation accuracy of 78.91% with a validation loss of 0.9607 and a training accuracy of 93.21% with a training loss of 0.1060. Strong convergence and effective feature learning are indicated by the minimal training loss, and the validation performance shows better generalization than convolutional neural network-based designs. The Vision Transformer's higher performance in multi-class retinal illness classification is attributed to its self-attention mechanism, which permits global contextual modelling across retinal fundus pictures. This allows the model to capture subtle and spatially distributed diseased signals.

IV. EXPERIMENTAL RESULTS

Using fundus images, the experimental research assesses how well different deep learning models perform for multi-class retinal illness categorization. A CNN, InceptionV3, MobileNetV2, ResNet50, and the suggested ViT are among the models used in this investigation. To provide a fair comparison, every model is trained using the same experimental setup. Accuracy and loss measures, which represent each architecture's learning efficiency and convergence behaviour, are used to evaluate model performance. Additionally, class-wise classification performance is examined using ROC curves and confusion matrices. is used to depict the prominent retinal areas impacting model predictions in order to improve model interpretability and provide explainable insights into the decision-making process.

Model-wise Accuracy Comparison

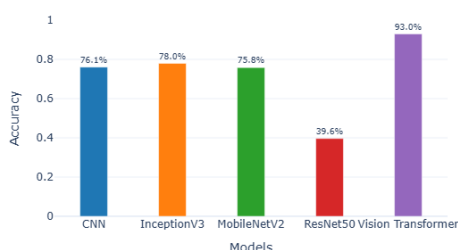


Fig 2. Model-wise Accuracy Comparison

Fig. 2 provides a comparison of the classification accuracy of various deep learning models used to classify retinal diseases. Moderate accuracy scores between roughly 75% and 78% are attained using conventional convolutional neural network-based models, such as CNN, InceptionV3, and MobileNetV2. Under the specified training configuration, ResNet50 exhibits noticeably decreased accuracy, suggesting limited adaption to the retinal illness dataset. The Vision Transformer, on the other hand, attains the maximum accuracy of 93.0%, demonstrating its exceptional capacity to model global contextual information and successfully differentiate between various retinal disease classifications.

Model-wise Loss Comparison

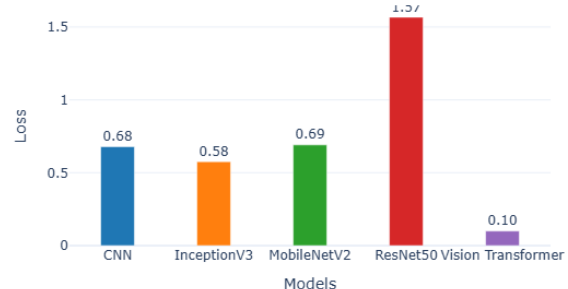


Fig 3. Model-wise Loss Comparison

Fig. 3 compares the loss values of many deep learning models that have been trained to classify retinal diseases. Partial convergence during training is indicated by the modest loss values of CNN, InceptionV3, and MobileNetV2, which fall between roughly 0.58 and 0.69. Due to weak convergence and limited dataset adaptability under the specified training setting, ResNet50 exhibits a significantly larger loss. When compared to convolutional neural network-based architectures, the Vision Transformer exhibits better optimization stability and more efficient learning of discriminative retinal characteristics, achieving the lowest loss value of 0.10.

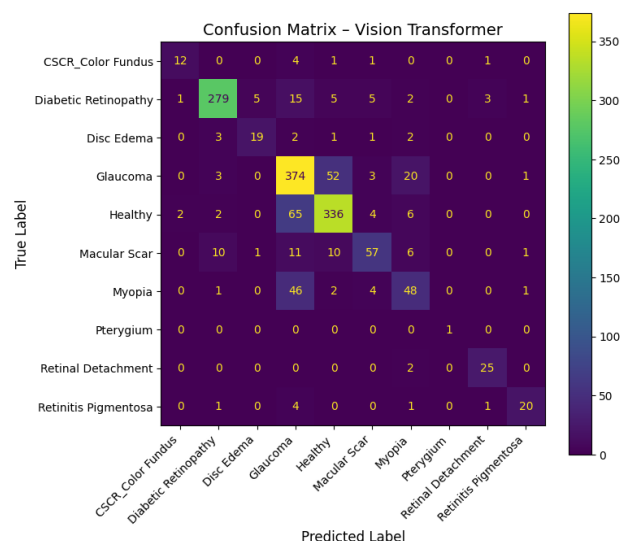


Fig. 4. Confusion Matrix of Vision Transformer

Fig.4. represents the Vision Transformer model's class-wise prediction ability for multi-class retinal illness categorization is displayed in the confusion matrix. Effective discrimination is demonstrated by the strong diagonal values, which show a high number of accurate predictions across the majority of disease categories. The robustness of the algorithm in recognizing clinically relevant illnesses is demonstrated by the exceptionally high true positive rates of diseases like Glaucoma, Healthy, Diabetic Retinopathy, and Retinal Detachment. Due to overlapping retinal features, there are few misclassifications between visually comparable classes, such as myopia, macular scar, and disc edema. Overall, the confusion matrix supports the Vision Transformer model's improved categorization consistency and dependability.

V. CONCLUSION

Conventional forms of convolution such as CNN, InceptionV3, and MobileNetV2 showed moderate learning ability, to acceptable accuracy and loss. values. These models are limited in that they cannot perform well, however. of complicated and worldwide retinal designs, especially in discriminating between ostensibly identical bodies of sickness. ResNet50 did not fit to the dataset with the mentioned. Poor convergence and resulting in poor convergence and training configuration. increasing loss. Instead, the ViT did better than the rest. of the tested models, having the greatest classification. accuracy and the lowest loss. The self-consciousness of the transformer. mechanism allowed successful modelling of international contextual. connection between retinal images, enhancing discrimination. through numerous classes of sickness. Furthermore, by identifying retinal areas which are clinically critical and affect predictions, Grad-CAM enhanced the interpretability of the model and enhancing transparency and trust of the diagnostics process. These results indicate that topologies based on transformers. possess a great potential of practical clinical screening. applications and especially availed of processing ocular. pictures. In order to enhance the scalability and reliability of diagnosis, In future studies, it can be considered to study larger datasets, multimodal. retinal imaging and real-time deployment.

VI. REFERENCES

- [1] A. Saini, K. Guleria, and S. Sharma, "An efficient deep learning model for eye disease classification," in 2023 International Research Conference on Smart Computing and Systems Engineering (SCSE), vol. 6, pp. 1–6, IEEE, 2023.
- [2] G. M. Niloy, A. K. Bitto, K. B. M. B. Biplob, M. H. Sammak, A. Das, and G. G. Hridoy, "MobileNet-Eye: An efficient transfer learning for eye disease classification," in 2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iACCESS), pp. 01–05, IEEE, 2024.
- [3] S. O. Purba, N. Khairina, R. Muliono, and A. H. Lubis, "Classification of Eye Diseases in Humans Using Vision Transformer Architecture Model," in 2024 International Conference on Information Technology Research and Innovation (ICITRI), pp. 71–75, IEEE, 2024.
- [4] K. Geethika, K. Jithi, and P. S. Rao, "Deep Learning for Eye Disease Classification: A Comparative Study of EfficientNet-B3 And ResNet-50," in 2025 International Conference on Sensors and Related Networks (SENNET) – Special Focus on Digital Healthcare (64220), pp. 1–6, IEEE, 2025.
- [5] R. E. Varghese and I. A. Pandian, "Inception-ResNetV2 based eye disease classification using retinal images," in 2023 3rd International Conference on Mobile Networks and Wireless Communications (ICMNWC), pp. 1–5, IEEE, 2023.
- [6] M. Giat and S. Hadi, "Convolutional Neural Network in Human Eye Disease Classification," in 2024 International Conference on Information Technology Systems and Innovation (ICITSI), pp. 348–353, IEEE, 2024.
- [7] H. A. Shah, S. Andberg, A. M. Koivisto, and R. Bednarik, "A Multimodal Approach for Early Identification of Mild Cognitive Impairment and Alzheimer's Disease With Fusion Network Using Eye Movements and Speech," IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2025.
- [8] M. A. Zain, T. A. B. Wirayuda, and F. Sthevanie, "Classification of Eye Diseases Using CNN on Fundus Images," in 2025 International Conference on Data Science and Its Applications (ICoDSA), pp. 44–48, IEEE, 2025.
- [9] P. Kumar, S. Bhandari, and V. Dutt, "Pre-Trained Deep Learning-Based Approaches for Eye Disease Detection," in 2023 International Conference on Circuit Power and Computing Technologies (ICCPCT), pp. 1286–1290, IEEE, 2023.
- [10] J. Orlosky, Y. Itoh, M. Ranchet, K. Kiyokawa, J. Morgan, and H. Devos, "Emulation of physician tasks in eye-tracked virtual reality for remote diagnosis of neurodegenerative disease," IEEE Transactions on Visualization and Computer Graphics, vol. 23, no. 4, pp. 1302–1311, 2017.
- [11] S. Wardani and P. Sihombing, "Hybrid of support vector machine algorithm and k-nearest neighbor algorithm to optimize the diagnosis of eye disease," in 2020 3rd International Conference on Mechanical, Electronics, Computer, and Industrial Technology (MECnIT), pp. 321–326, IEEE, 2020.
- [12] Y. Li, P. W. Chiu, V. Tam, A. Lee, and E. Y. Lam, "Dual-mode imaging system for early detection and monitoring of ocular surface diseases," IEEE Transactions on Biomedical Circuits and Systems, vol. 18, no. 4, pp. 783–798, 2024.
- [13] S. K. Shinde, A. Buchade, P. Mahalle, M. Vengurlekar, S. Yedge, and G. Arsalwad, "Eyes on Health: A Deep Learning Approach for Ocular or Eyes Disease Detection," in 2024 IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS), pp. 1–5, IEEE, 2024.
- [14] V. T. T. Huong, T. T. H. Long, and H. Q. Tran, "Identification and Diagnosis of Eye Diseases Using Deep Learning and YOLO," in 2025 8th International Conference on Circuits, Systems and Simulation (ICCS), pp. 70–77, IEEE, 2025.
- [15] Kumar, K. Antony, MJ Carmel Mary Belinda, V. Dhillip Kumar, Jerlin Francy Rajan, and Muhammad Arif. "A nonlinear recurrent encoders for early detection of strep throat infection to prevent acute rheumatic fever." *Journal of Ambient Intelligence and Humanized Computing* 15, no. 4 (2024): 2201-2213.